Motion Planning: Formal Definition and Plain RRT Pseudocode

1 Path Planning Problem

Given a configuration space $\mathcal{C}$ and obstacle region $\mathcal{C}_{\text{obs}} \subseteq \mathcal{C}$, and start and goal $q_{\text{start}}, q_{\text{goal}} \in \mathcal{C}_{\text{free}}$, where $\mathcal{C}_{\text{free}} = \mathcal{C} \setminus \mathcal{C}_{\text{obs}}$, find a continuous collision-free path $\sigma : [0,1] \to \mathcal{C}_{\text{free}}$ satisfying

$$\sigma(0) = q_{\text{start}},$$
$$\sigma(1) = q_{\text{goal}}.$$

For a goal region $\mathcal{G}$, require instead $\sigma(1) \in \mathcal{G} \subseteq \mathcal{C}_{\text{free}}$.

Algorithm 1 Rapidly-exploring Random Tree (RRT)

function RRT($q_{\text{start}}, q_{\text{goal}}, \Delta$)
  $T \leftarrow \{q_{\text{start}}\}$
  repeat
    $q_{\text{rand}} \leftarrow$ RandomPointInMap
    $q_{\text{near}} \leftarrow$ Nearest(T, q_{rand})
    $q_{\text{new}} \leftarrow$ Steer($q_{\text{near}}, q_{\text{rand}}, \Delta$)
    if $\neg$CollisionFree($q_{\text{near}}, q_{\text{new}}$) **then**
      continue
    end if
    $T \leftarrow T \cup \{q_{\text{new}}\}$
    if $\|q_{\text{new}} - q_{\text{goal}}\| \leq \Delta$ and CollisionFree($q_{\text{new}}, q_{\text{goal}}$) **then**
      $T \leftarrow T \cup \{q_{\text{goal}}\}$
      return ReconstructPath(T, parent, q_{goal})
    end if
  until false
end function